DUROTECH

# DURO CLAY LINER

## GEOSYNTHETIC BENTONITE LINER

Duro Clay Liner is a laminated composite sheet membrane containing bentonite clay. Bentonite is a natural inert clay, formed from volcanic ash, that absorbs water. It can swell to form an impermeable mass that prevents the passage of liquid and provides waterproofing properties. Fibres from the non-woven geotextile fabric are needle punched through the layer of bentonite to ensure there is no movement of the clay in handling and cutting around penetrations. Through its ability to swell it can withstand minor movement and all joints are self sealing at the overlap.This product has been developed for use in basement weatherproofing, sandwich-slab construction and lift shafts.

## APPLICATION

- Duro Clay Liner can be laid by hand or me-chanically.
- When laying the liner, the seam overlaps must be staggered.
- All laps should be free from sand, soil or other impurities that could effect the quality of the join.
- When fixing to a sloping or vertical position, it is preferable to mechanically fix the liner in place and withdraw fixings progressively when backfilling or cover fixings with sealant.
- The Duro Clay Liner is manufactured with two types of geotextile fabric (one to each face). The woven geotextile fleece must be placed against the structure.
- The subgrade must be compacted before laying the Duro Clay Liner, and the surface should be smooth with minimal indentation.
- The surface should be preferably dry, how-ever damp conditions can be catered for.
- Minimum clearance from excavation should be 600mm or as directed by local safety re-quirements, and on sloping sites the angle of batter should not exceed 30 degrees.
- It is important to establish a safe, clean site prior to commencing work.
- The liner must be backfilled the same day so as to protect the membrane's integrity.
- Ideally backfill with sand or aggregate to a maximum of 10mm and compact to a mini-mum of 85% dry density in layers of 150mm.
- Do not damage the membrane when backfilling. Use protection board or drainage cell.



## TECHNICAL SPECIFICATION

| Material Property | Test Result |
|---|---|
| Mass per unit area of bentonite mat | 6015gm/m$^2$ |
| Thickness of ben-tonite mat | 6.5mm |
| Size | 1.1mtrs x 20mtrs |
| Tensile strength | 11.2KN/m$^2$ |
| Adhesive strength | 68cm |
| California bearing ratio | 1.16KN |
| Permeability | 8.64X10-10cm/a |
| Expansion coefficient of bentonite | 26.3ml/2g |
| Moisture content of bentonite | 12% |
| Loss of liquidated bentonite | 14.7ml |

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*The information provided in this data sheet is correct at the time of printing (albeit is subject to change at any time) and is intended to give a simple description of the product and its capabilities. In practice, the substrate, intended surface to be treated and environmental conditions vary widely, making it essential for the user to determine the products suitability for a particular application and to ensure that the product is not used beyond its physical limitations. If in doubt contact the manufacturer. The product will perform as described herein provided it is applied in accordance with the manufacturer's instructions as stated in this data sheet and provided that the building and installation is structurally sound and the application is carried out competently. Durotech terms and conditions of sale apply.*